To: Sarah
From: Kanchan Patkar
Subject: GiftLane — Customer Segmentation and Product Recommendation Results

Dear Sarah,

I have completed the customer segmentation and recommendation-model sensitivity analysis for GiftLane, an e-commerce analytics project using the Online Retail II dataset. The project aims to understand customer purchasing behavior and develop a data-driven approach to product recommendations.

The customer segmentation analysis mapped 5,483 customers into five segments. Each segment was profiled using its most frequently purchased products and leading countries. The United Kingdom was the leading country across all five reported profiles, while the popular products varied between groups. The largest segment contained 4,260 customers, indicating that the segment sizes were highly imbalanced. These findings provide an initial basis for exploring customer purchasing patterns and segment-specific recommendations, although further validation is needed to assess the business usefulness of the segments.

I also evaluated three purchase-value representations—Quantity, Binary (1/0), and Log Quantity—using Truncated SVD with 200 components. All models were evaluated on the same 1,913 eligible customers and held-out test targets using Hit Rate@10. Raw Quantity achieved 25.51%, Binary achieved 39.47%, and Log Quantity achieved the highest score of 39.78%. The Log Quantity model produced at least one relevant held-out product in the top 10 recommendations for 761 customers.

These results indicate that raw purchase quantities performed worse than binary and log-transformed representations. Log Quantity was the best-performing option among the three tested, although its improvement over Binary was only 0.31 percentage points. The measured Log Quantity fitting time was 2.8291 seconds, and its evaluation time was 0.7051 seconds. These timings cover the measured operations, not the complete nightly pipeline.

Based on these findings, I recommend carrying Log Quantity forward for additional testing rather than treating it as a finalized production choice. The next steps are to compare the recommendation model against the top-10 best-seller baseline, validate customer segment quality, and measure end-to-end processing time and storage requirements.

Overall, GiftLane demonstrates a workflow for preparing retail transaction data, profiling customer segments, and evaluating recommendation approaches using a time-based holdout. Further benchmarking and validation will help determine its practical value for retail personalization.

Sincerely,
Kanchan Patkar
GiftLane Individual Project